

ASSIGNMENT REPORT

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COURSE TITLE: COMPUTER PROGRAMMING II (PYTHON)

COURSE CODE: COS 202

DEPARTMENT: INFORMATION AND COMMUNICATION TECHNOLOGY

Development of a Mathematical Calculator and Personal Pocket CGPA Calculator Using Python

Introduction

This project involves the development of two simple Python applications: a Mathematical Calculator (MC) and a Personal Pocket CGPA Calculator (PPC). The aim is to apply Python programming concepts such as variables, input/output, arithmetic operators, loops, and selection control statements (if, elif, and else) to solve practical problems.

Objectives

- To develop a simple Mathematical Calculator for performing basic arithmetic operations.
- To develop a Personal Pocket CGPA Calculator for calculating students' CGPA.
- To improve programming skills using the Python programming language.

Project Description

1. Mathematical Calculator (MC)

The Mathematical Calculator is a menu-driven Python program that allows users to perform basic arithmetic operations. The calculator supports addition, subtraction, multiplication, division, floor division, exponentiation, modulus, clear screen, and exit functions.

2. Personal Pocket CGPA Calculator (PPC)

The Personal Pocket CGPA Calculator allows users to enter course units and scores. The program determines the grade and grade point for each course, calculates the total quality points, computes the CGPA, and displays the appropriate degree classification.

Tools Used

- Python 3
- Pydroid 3 (Python IDE for Android)

Features

- Interactive menu
- User-friendly interface
- Error checking for division by zero
- Automatic CGPA calculation
- Automatic grade and degree classification

Conclusion

The project demonstrates the practical application of Python programming in solving everyday computational problems. It also improves understanding of arithmetic operations, control statements, loops, and user interaction in Python.